

# PX-100 Industrial Circulation Pump

Operating and Maintenance Manual - Revision 3.1

This is a fictional document created for AI testing training. The product, values and procedures are invented. Do not use for real equipment.

## 1. General safety notes

- Read this manual completely before installing, operating or servicing the pump.
- Only trained personnel may carry out installation and maintenance work.
- Wear safety glasses, gloves and hearing protection when working near the pump in operation.
- Keep the area around the pump clear of tools and loose material.

## 2. Product overview

The PX-100 is a compact centrifugal pump for closed-loop water circulation in small production lines. It is designed for continuous operation at moderate pressure.

## 3. PX-100 operating limits

Operate the PX-100 only within the limits below. Operating outside these limits can damage the pump and void any service agreement.

| Parameter                  | Limit                 |
|----------------------------|-----------------------|
| Maximum operating pressure | 10 bar                |
| Fluid temperature range    | 20 - 40 °C            |
| Maximum flow rate          | 45 m3/h               |
| Motor power                | 5.5 kW                |
| Supply voltage             | 400 V, 3-phase, 50 Hz |

Do not bypass or disable the pressure interlock under any circumstances.

## 4. Installation

- Mount the pump on a level, vibration-free base.
- Align the pump and motor shafts before first start-up.
- Ensure at least 30 cm of free space around the motor for ventilation.
- Fill the pump casing with fluid before starting. Never run the pump dry.

## 5. Preventive maintenance

Preventive maintenance for the PX-100 is required every 6 months. Record each maintenance visit in the equipment log.

| Task                                | Interval        |
|-------------------------------------|-----------------|
| Inspect seals for leakage           | 6 months        |
| Check bearing temperature and noise | 6 months        |
| Replace lubricant                   | Every 12 months |
| Check shaft alignment               | Every 12 months |

## 6. Preparing the pump for maintenance

1. Switch off the pump and isolate the electrical supply at the main switch.
2. Apply lock-out/tag-out to the main switch so the pump cannot be restarted.
3. Close the inlet and outlet valves.
4. Release the pressure from the pump casing using the drain valve, and confirm the gauge reads 0 bar.
5. Allow the pump to cool to below 40 °C before touching the casing.
6. Only then open the casing or inspect the impeller. Never inspect the impeller while the pump is running.

## 7. Troubleshooting: overheating

| Possible cause                           | Action                                    |
|------------------------------------------|-------------------------------------------|
| Blocked cooling path or poor ventilation | Clear obstructions; check 30 cm clearance |
| Low or degraded lubricant                | Replace lubricant (see section 5)         |
| Running above rated pressure             | Reduce load; do not exceed 10 bar         |
| Worn bearings                            | Inspect bearings; replace if noisy or hot |
| Shaft misalignment                       | Re-align pump and motor shafts            |

If the pump overheats repeatedly, stop operation and raise a service request with your maintenance team.

## 8. Contact

For service requests, contact your site maintenance team using the internal service request process.